SYLLABUS

B.TECH.

COMPUTER SCIENCE & ENGINEERING

Under

Choice Based Credit System (CBCS)

Head of the Department
Computer Engineering & Applications
Institute of Engineering & Technology
GLA University, Mathura

BCSC1002: OBJECT ORIENTED PROGRAMMING

Objective: *This course introduces the Object-Oriented programming paradigm to students. It also teaches a student how to think objectively and model a Java program for solving real-world problems.*

CREDITS: 3 **L-T-P-J:3-0-0-0**

Module No.	Content	Teaching Hours
I	**Object-Oriented Programming:** Features of Object-Oriented Programming, Introduction to Object-Oriented Java Programming. **g Java Technology & Environment:** Understanding the compilation process of the JVM, JVM vs JDK vs JRE, Key Features of Java, Structure of a simple Java program. **Working with Java Primitive Data Types:** Strongly Typed nature of Java, Primitive Data Types in Java, The new 'var' keyword, Scope of a variable. **Accepting User Input in Java Programs:** using the Scanner class, using command line arguments. **Programming Constructs:** Sequence, Selection, Iteration & Transfer Statements, For-Each Loop. **Working with Java Arrays**: Declaring and Initializing One-Dimensional and Two-Dimensional Arrays in Java, Introduction to java. util. Arrays class. **The String API:** String Data Type, commonly used methods from the String API, String Tokenizer, String Builder & String Buffer. **Creating and Using Methods:** Signature of a method, Types of Methods, Overloading methods in a class, Static and Non-Static Methods. **Describing and Using Objects & Classes:** Declare the structure of a Java class, declaring members of a class (fields and methods), declaring and using Java Objects, lifecycle of an Object (creation, assignment, dereferencing and garbage collection), Constructors of a class, Overloading Constructors, Constructor chaining using 'this' and 'super' keyword. **Using Java Packages:** create and import Java packages and static imports, abstracting program logic to packages, creating executable main class, running the executable class inside a package. **Applying Encapsulation**: Using access modifiers with/in a class, principles of encapsulation. **Programming Abstractly Through Interfaces:** create and implement Interfaces for programs, private and default methods in Interfaces, declaring Abstract Classes, Constructors in Abstract Classes. Marker Interface, Functional Interfaces, Lambda Expressions in Java.	20
II	**Reusing Implementations using Inheritance:** Declaring Subclasses and Super classes, extend Abstract Classes, implementing Interfaces, exploring polymorphic behavior by overriding methods, Object Types vs Reference Types, differentiate overloading, overriding and hiding. **Exception Handling:** Exception Hierarchy, Need of Exception Handling, Checked Exceptions, Unchecked Exceptions and Errors, Try-Catch Blocks, Finally, Throw & Throws Keywords, creating and handling Custom Exceptions. **Threads in Java:** Life Cycle of a Thread, creating threads using Runnable and Thread, 'sleep ()', Thread Priorities. **Using Wrapper Classes:** Wrapper Classes in Java, Boxing-Unboxing-Auto Boxing-Auto Unboxing. **Generics & Collections:** Creating Generic classes, Generic Methods, Diamond Notation, Wildcards, Type Erasure, Collection Hierarchy, Base Interfaces, Lists, Sets and Maps. **The Stream API:** Introduction to the Stream API, using lambda expressions in Streams. **Regular Expressions:** Pattern and Matcher Class. **JDBC:** JDBC Drivers, Connecting to a MySQL Database, Driver Manager, Connection Interface, Statement Interface, Result Set Interface, Prepared Statements.	18

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Text Book:

- Herbert Schildt, "The Complete Reference, Java Eleventh Edition", Oracle Press.2019.

Reference Book:

- Cay S Hosrtmann, "Core Java Volume I—Fundamentals, Eleventh Edition", Pearson,2018.
- Rogers Cadenhead, "Sams Teach Yourself Java in 21 Days (Covers Java 11/12), 8th Edition", Pearson,2020.

Outcomes: After completion of the course, students will be able to -

- CO1: Understand the basics of Object-Oriented Programming paradigm.
- CO2: Construct the logical flow of programs by using the sequence, selection, iterations and transfer statements.
- CO3: Apply the concepts of Object-Oriented Programming to model programs in Classes, Abstract Classes, Interfaces and Enums, and simplify program function by dissecting it into methods.
- CO4: Understand accessibility of members in a program unit and create packages to prevent namespace collisions.
- CO5: Predict run-time errors in a program by examining program functioning.
- CO6: Show the parallel processing capabilities of a program using a multithreading concept.
- CO7: Experiment with the predefined classes and interfaces defined in the Collections Framework.
- CO8: Develop a program using JDBC connectivity to demonstrate data persistence.

Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs):

COs	POs/PSOs
CO1	PO1,PO3/PSO1,PSO2
CO2	PO1,PO3/PSO1,PSO2
CO3	PO1,PO2/PSO1,PSO2
CO4	PO1/PSO2,PSO4
CO5	PO1,PO2,PO4/PSO4
CO6	PO1,PO2, PO3/PSO2
CO7	PO1,PO2,PO11/PSO2
CO8	PO1,PO2,PO3/PSO1,PSO2

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BCSC1003: DATABASE MANAGEMENT SYSTEM

Objective: *The objective of the course is to enable students to understand and use a relational database& NoSQL system. Students learn how to design and create a good database.*

Credits: 03 **L-T-P-J: 3-0-0-0**

Module No.	Content	Teaching Hours
I	**Introduction:** An Overview of Database Management System, Database System Vs File System, Database System Concept and Architecture, Data Model Schema and Instances, Data Independence, Database Language and Interfaces (DDL, DML, DCL), Database Development Life Cycle (DDLC) with Case Studies. **Data Modeling Using the Entity-Relationship Model:** ER Model Concepts, Notation for ER Diagram, Mapping Constraints, Keys, Specialization, Generalization, Aggregation, Reduction of an ER Diagram to Tables, Extended ER Model. **Relational Data Model and Language:** Relational Data Model Concepts, Integrity Constraints, Entity Integrity, Referential Integrity, Keys Constraints, Primary Key, Foreign Key, Candidate Key, Super Key, Domain Constraints, Relational Algebra. **Database Design & Normalization I:** Functional Dependencies, Canonical Cover, Normal Forms, First, Second, Third Normal Forms, BCNF.	20
II	**Database Design & Normalization II:** Lossless Join and Dependency Preserving Decomposition, MVD and 4th Normal Form, JD and 5th Normal Form, Inclusion Dependence. **File Organization:** Indexing, Structure of Index files and Types, Dense and Sparse Indexing **Transaction Processing Concept:** Transaction System, Testing of Serializability, Serializability of Schedules, Conflict & View Serializable Schedule, Recoverability, Recovery from Transaction Failures, Log Based Recovery, Deadlock Handling. **Concurrency Control Techniques:** Concurrency Control, Locking Techniques for Concurrency Control, 2PL, Time Stamping Protocols for Concurrency Control, Validation Based Protocol. **Distributed Database:** Introduction of Distributed Database, Data Fragmentation and Replication.	20

Text Books:

- Elmasri and Navathe, "Fundamentals of Database Systems", 6th Edition, Addison Wesley,2010.
- Sadalage, P. &Fowler, "NoSQL Distilled: A Brief Guide to the Emerging World of Polyglot Persistence", Pearson Education,2012.

References Books:

- Date C J," An Introduction to Database Systems", 8th Edition, Addison Wesley.
- Korth, Silbertz and Sudarshan, "Database Concepts", 5th Edition, TMH,1998.
- Redmond, E. & Wilson, "Seven Databases in Seven Weeks: A Guide to Modern Databases and the NoSQL Movement", 1st Edition.

Outcome: After the completion of the course, the student will:

- CO1: Understand the concept of database management systems and Relational database.
- CO2: Identify the various data model used in database design.
- CO3: Design conceptual models of a database using ER modeling for real life applications and construct queries in Relational Algebra.
- CO4: Create and populate a RDBMS for a real life application, with constraints and keys using SQL.
- CO5: Select the information from a database by formulating complex queries in SQL.
- CO6: Analyze the existing design of a database schema and apply concepts of normalization to design an optimal database.
- CO7: Discuss indexing mechanisms for efficient retrieval of information from a database.
- CO8: Discuss recovery system and be familiar with introduction to web database, distributed databases.

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Mapping of Course Outcomes (COs) withProgram Outcomes (POs) and Program Specific Outcomes (PSOs):

COs	POs/PSOs
CO1	PO1 /PSO1
CO2	PO2, PO3/ PSO2
CO3	PO2,PO3,PO6,PO11/PSO1,PSO2,PSO4
CO4	PO1,PO3/PSO1
CO5	PO1,PO5/PSO1
CO6	PO2,PO3,PO9/ PSO2
CO7	PO1,PO11 /PSO1
CO8	PO1,PO3,PO12/ PSO2

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BCSC0004: OPERATING SYSTEMS

Objective: *This course aims to introducing the concept of computer organization. In particular, it focuses on basic hardware architectural issues that affect the nature and performance of software.*

Credits:03 **L-T-P-J:3-0-0-0**

Module No.	Content	Teaching Hours
I	**Introduction:** Operating System and its Classification - Batch, Interactive, Multiprogramming, Time sharing, Real Time System, Multiprocessor Systems, Multithreaded Systems, System Protection, System Calls, Reentrant Kernels, Operating System Structure- Layered structure, Monolithic and Microkernel Systems, Operating System Components, Operating System Functions and Services. **Processes:** Process Concept, Process States, Process State Transition Diagram, Process Control Block (PCB), Process Scheduling Concepts, Threads and their management. **CPU Scheduling:** Scheduling Concepts, Performance Criteria, Scheduling Algorithms, Multiprocessor Scheduling. **Process Synchronization:** Principle of Concurrency, Implementation of concurrency through fork/join and parbegin/parend, Inter Process Communication models and Schemes, Producer / Consumer Problem, Critical Section Problem, Dekker's solution, Peterson's solution, Semaphores, Synchronization Hardware. **Classical Problem in Concurrency:** Dining Philosopher Problem, Readers Writers Problem.	20
II	**Deadlock:** System model, Deadlock characterization, Prevention, Avoidance and detection, Recovery from deadlock, Combined Approach. **Memory Management:** Multiprogramming with fixed partitions, Multiprogramming with variable partitions, Paging, Segmentation, Paged segmentation. **Virtual memory concepts:** Demand paging, Performance of demand paging, Page replacement algorithms, Thrashing, Locality of reference. **I/O Management and Disk Scheduling:** I/O devices, I/O subsystems, I/O buffering, Disk storage and disk scheduling. **File System:** File concept, File organization and access mechanism, File directories, File allocation methods, Free space management.	20

Text Books:

- Silberschatz, Galvin and Gagne, "Operating Systems Concepts",9th Edition, Wiley, 2012.

Reference Books:

- SibsankarHalder and Alex a Aravind," Operating Systems", 6th Edition, Pearson Education, 2009.
- Harvey M Dietel, "An Introduction to Operating System", 2nd Edition, Pearson Education, 2002.
- D M Dhamdhere, "Operating Systems: A Concept Based Approach", 2nd Edition, 2006.
- M. J. Bach, "Design of the Unix Operating System", PHI, 1986.

Outcome: After completion of course, the student will be able to:

- CO1: Understand the classification of operating system environment.
- CO2: Understand the basic of process management.
- CO3: Apply the concept of CPU process scheduling for the given scenarios.
- CO4: Illustrate theprocess synchronization and concurrency process in operating system.
- CO5: Analyze the occurrence of deadlock in operating system.
- CO6: Describe and analyze the memory management and its allocation policies.
- CO7: Understand the concepts of disk scheduling.

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Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs):

COs	POs/PSOs
CO1	PO1,PO2,PO7/PSO1
CO2	PO1,PO2 /PSO1
CO3	PO1,PO4/PSO1,POS3
CO4	PO3,PO4,PO6/PSO3,PSO4
CO5	PO1,PO4/PSO1,PSO3
CO6	PO1,PO2/PSO1,PSO3
CO7	PO1,PO2,PO7/PSO1,PSO3

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BCSC1005: COMPUTER ORGANIZATION

Objective: *This course aims to introducing the concept of computer organization. In particular, it focuses on basic hardware architectural issues that affect the nature and performance of software.*

Credits: 03 **L–T–P-J: 3–0–0-0**

Module No.	Content	Teaching Hours
I	**Basic Organization:** Basic organization of the computer and Block level description of the functional units, Number representation; 1's and 2's Complement, Integer Representation, Arithmetic Addition & Subtraction with overflow. fixed and floating-point number representation, IEEE standard floating point representation. Introduction to Combinational Circuit- half adder, full adder, binary adder/subtractor, carry look ahead adders. Multiplexer and Demultiplexer, Register, bus and memory transfer. **Central Processing Unit:** Addition and subtraction of signed numbers, Multiplication: Signed operand multiplication, Booths algorithm. Processor organization, general registers organization, stack organization, Three, Two, One & Zero address instruction. Addressing modes, Micro-operations (Arithmetic, Logical & Shift) and its applications.	20
II	**Multiprogramming and Multiprocessing:** Flynn's classification, Introduction to pipelined operation. Instruction types, formats, Instruction cycles. **Control Unit:** Execution of a complete instruction. Hardwired and micro programmed control unit. Unconditional and Conditional branching. Microinstruction with next address field, pre-fetching microinstructions, Concept of horizontal and vertical microprogramming. **Memory:** Basic concept of Memory and its hierarchy, RAM memories, 2D, 2 & 1/2D memory organization. ROM memories. Cache memories: concept and design issues, performance, address mapping and replacement. Virtual memory: concept and implementation. **Input/Output:** Peripheral devices, I/O interface, I/O ports, Interrupts: interrupt hardware, types of interrupts and exceptions. Buses, bus architecture, types of buses and bus arbitration. Modes of Data Transfer: Programmed I/O, interrupt initiated I/O and Direct Memory Access., I/O channels and processors. Standard communication interfaces.	**20**

Text Books:

- M. Mano, "Computer System Architecture", 3rd Edition, PHI,1996

Reference Books:

- D.W. Patterson, "Computer Organization and Design", 4th **Edition**, Elsevier Publication, 2008.
- William Stalling, "Computer Organization",8th Edition, PHI, 2011.
- V. CarlHamacher, Zaky, "Computer Organization",4th International Edition, TMH, 1996.
- John P Hays, "Computer Organization", 2nd Edition, TMH.
- Tannenbaum , "Structured Computer Organization", 5thEdition, PHI, 2005.
- P Pal Chaudhry, "Computer Organization & Design", 2ndEdition, PHI, 2002.

Outcome: After completion of the course, the student will be able to:

- CO1: Understand the basics of digital computer system.
- CO2: Demonstrate the principle of arithmetic operations on unsigned, signed integers and floating point numbers.
- CO3: Understand the concepts of Combinational and Sequential circuits and their applications.
- CO4: Understand the CPU architecture and organization.
- CO5: Explain the basic concepts of pipelining.

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- CO6: Design the steps for the execution of the complete instruction for hardwired and micro-programmed control unit.
- CO7: Explain the function of memory hierarchy.
- CO8: Determine the interface of CPU with input/output devices and their modes of transfer.

Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs):

COs	POs/PSOs
CO1	PO1,PO3/PSO1
CO2	PO1,PO3/PSO1
CO3	PO2,PO3,PO5/PSO2
CO4	PO2,PO3,PO4/PSO1,PSO3
CO5	PO2,PO3,PO4/PSO2
CO6	PO1,PO2,PO3/PSO1,PSO3
CO7	PO2,PO3,PO5/PSO2,PSO3
CO8	PO3,PO4/PSO1

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BCSC0006: DATA STRUCTURES AND ALGORITHMS

Objective: *The objective of this course is that students will construct and application of various data structures and abstract data types including lists, stacks, queues, trees and graphs.*

Credits: 04 **L–T–P-J: 3–1–0-0**

Module No.	Content	Teaching Hours
I	**Introduction:** Basic Terminology, Elementary Data Organization, Properties of an Algorithm, Efficiency of an Algorithm, Time and Space Complexity, Asymptotic Notations – Big-Oh; Operations on Data Structure, Abstract Data Types (ADT). **Linked Lists:** Implementation of Singly Linked Lists, Doubly Linked List, Circular Linked List, Operations on a Linked List - Insertion, Deletion, Traversal; Generalized Linked List, Polynomial Representation and Addition. **Stacks:** Primitive Stack Operations - Push & Pop, Array and Linked Implementation of Stack in C, Application of Stack: Prefix and Postfix Expressions, Evaluation of Postfix Expression, conversion of Infix to Postfix expression, Recursion, Principles of Recursion, Tail Recursion, Removal of Recursion, use of stack in Recursion, Tower of Hanoi Problem. **Queues:** Operations on Queue - Add, Delete operations, Implementation of Queue Using Array and Linked List, Circular Queues, Deque and Priority Queue. **Trees:** Basic Terminology, Array Representation and Dynamic Representation; Complete Binary Tree, Algebraic Expressions, Extended Binary Trees, Tree Traversal Algorithms - Inorder, Preorder and Postorder; Threaded Binary Trees, Traversing Threaded Binary Trees.	20
II	**Search Trees:** Binary Search Trees (BST), Insertion and Deletion in BST, AVL Trees, Introduction to M-Way Search Trees, B Trees. **Searching:** Sequential Search, Binary Search. **Sorting:** Bubble Sort, Selection Sort, Insertion Sort, Quick Sort, Two Way Merge Sort, and Heap Sort. **Graphs:** Terminology, Adjacency Matrices, Adjacency List, Graph Traversal - Depth First Search and Breadth First Search; Spanning Trees, Minimum Cost Spanning Trees – Prim's and Kruskal's Algorithm; Shortest Path Algorithm – Bellman-Ford and Dijkstra's Algorithm. **Hashing & Indexing:** Hash Function, Collision Resolution Strategies. Primary Indices, Secondary Indices, Indexing and Hashing Comparisons.	20

Text Book:

- Aaron M. Tanenbaum, YedidyahLangsam and Moshe J. Augenstein, "Data Structures Using C and C++", 2nd Edition, PHI, 2009.

Reference Books:

- Horowitz and Sahani, "Fundamentals of Data Structures", 3rd Edition, W H Freeman & Co, 2004-05.
- Jean Paul Trembley and Paul G. Sorenson, "An Introduction to Data Structures with Applications", 2nd Edition, TMH, 2007.
- R. Kruse, "Data Structures and Program Design in C", 2nd Edition, Pearson Education, 2004.
- Lipschutz Schaum's Outline Series, "Data Structures", 12th Reprint, TMH, 2010.
- G A V Pai, "Data Structures and Algorithms", TMH, 2009.

Outcome: After completion of course, student will be able to:

- CO1: Understand the basic concepts of the data structure and algorithms.
- CO2: Understand the complexity representation in terms of Big Oh, Theta and Omega notations.
- CO3: Apply the associated operations in linear data structure like stack, Queue and link list.
- CO4: Apply the associated operations in Binary Search Tree, AVL Tree and M- Way Search Tree.
- CO5: Understand the basic algorithms such as heap sort, graph traversal, quick sort, AVL trees, and hashing.
- CO6: Select the appropriate data structure to solve the problem.
- CO7: Apply the shortest path algorithm to solve real life problem.

Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs):

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COs	POs/PSOs
CO1	PO1/PSO1,PSO2
CO2	PO1, PO2/PSO1,PSO2
CO3	PO1/PSO1
CO4	PO1,PO4/PSO1
CO5	PO1,PO4/PSO3
CO6	PO2/PSO4
CO7	PO2/PSO4

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BCSC0007: INTRODUCTION TO MICROPROCESSORS

Objective: *Objective of this subject is to introduce the basic concepts of microprocessor and assembly language programming. Identify and explain the operation of the components of typical microprocessor: the role of the ALU, registers, stack and the use of interrupts.*

Credits: 03 **L–T–P-J: 3–0–0-0**

Module No.	Content	Teaching Hours
I	**Introduction**: Microprocessors Evolution and Types, Basics of Pentium Microprocessor, Microprocessor Application, **8-Bit Microprocessor:**8085 Microprocessor and its Architecture, Addressing Modes, The 8085 Programming Model, Instruction Classification, Instruction Format, Overview of Instruction Set - Data Transfer Operation, Arithmetic Operation, Logic Operations and Branch Operations; Introduction to Assembly Language Program. **Programming Technique with Additional Instruction**: Looping, Counting, Indexing, Additional Data Transfer and 16-Bit Arithmetic Instruction, Counters and Time Delays, Stack and Subroutine.	20
II	**16 Bit Microprocessor:** Architecture of 8086 – Register Organization, Execution Unit, Bus Interface Unit, Signal Description, Physical Memory Organization, Mode of Operation, I/O Addressing Capabilities. **Peripheral Interfacing:** I/O Programming, Programmed I/O, Interrupt Driven I/O, DMA I/O, Memory-Mapped I/Os. **Peripheral Devices:** 8237 DMA Controller, 8255 Programmable Peripheral Interface, 8253/8254 Programmable Timer/Counter, 8259 Programmable Interrupt Controller.	18

Text Books:

- N Senthil Kumar, MSaravanan, and S Jeevananthan, "Microprocessors and Microcontrollers", Oxford University Press India, 2010.

Reference Books:

- Ramesh S. Gaonkar , "Microprocessor Architecture Programming and Applications with 8085", 4th Edition, Penram International Publishing, 2000.
- Ray A.K. Bhurchandi.K.M, "Advanced Microprocessor and Peripherals", TMH, 2002.
- D. V. Hall, "Microprocessors and Interfacing: Programming and Hardware", 2nd Edition, TMH, 1992.
- Y.C. Liu and G.A. Gibson, "Microcomputer Systems: The 8086/8088 Family Architecture Programming and Design", 2nd Edition, PHI, 2003.

Outcome: After the completion of the course, the student will be able to:

- CO1: Demonstrate the Microprocessor internal architecture and its operations.
- CO2: Develop programs based on 8085 microprocessor instruction set and addressing mode.
- CO3: Develop program using looping, counting, indexing, counter and time delays.
- CO4: Understand the concept of stack and subroutine for modular approach.
- CO5: Compare accepted standards and guidelines to select microprocessor (8085 & 8086) to meet performance requirements.
- CO6: Analyze the concept of interfacing the processor to external device with I/O programming & Interrupt Driven I/O.
- CO7: Understand the working of interfacing chips (8237, 8253/54, 8255 & 8259).

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Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs):

COs	POs/PSOs
CO1	PO1,PO2/PSO1
CO2	PO2,PO3/PSO1,PSO2
CO3	PO2,PO3/PSO1,PSO2
CO4	PO1,PO2,PO3/PSO1,PSO3
CO5	PO2,PO3,PO5/PSO1,PSO3
CO6	PO1,PO2/PSO3
CO7	PO1,PO2,PO4/PSO3

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BCSC 0008: Computer Networks

Objective: *The objective is to understand fundamental underlying principles of computer networking, details and functionality of layered network architecture.*

Credits: 03 **Semester - IV** **L–T–P-J: 3–1–0-0**

Module No.	Content	Teaching Hours
I	**Introduction Concepts**: Goals and Applications of Networks, Network structure and architecture, The OSI reference model, services, Network Topology Design, Physical Layer Transmission Media, Line coding scheme, switching methods (circuit switching, Packet switching), TDM. **Medium Access sub layer**: Medium Access sub layer - Channel Allocations, LAN protocols - ALOHA protocols, CSMA, CSMA/CD, Overview of IEEE standards. **Data Link Layer**: Error detection and correction, Flow control (sliding window protocol)	20
II	**Network Layer**: Network Layer –IP addressing, subnet, CIDR, VLSM, Internetworking, Address mapping, routing. Connecting devices. **Transport Layer**: Transport Layer - Design issues, connection management, Flow control, TCP window management, congestion control-slow start algorithm. **Application Layer**: Data compression, Data Encryption, File Transfer, DNS, HTTP, SMTP, TELNET **Introduction to IPv6, transition from IPv4 to IPv6.**	20

Text Books:

- Forouzan B. A. , "Data Communication and Networking", 4th Edition, McGrawHill,2004.

References:

- Kurose, J.F. and Ross K.W., "Computer Networking: A Top-Down Approach Featuring the Internet", 3rd Edition, Addison-Wesley,2005.
- A.S. Tanenbaum, "Computer Networks", 2nd Edition, Prentice Hall India,2006.

Outcome: After the completion of the course, the student will be able to:

- CO1: Understand the concept of OSI and TCP/IP reference model.
- CO2: Understand the basics of data transmission at physical layer.
- CO3: Understand the channel allocation using ALOHA, CSMA and CSMA/CD.
- CO4: Apply error detection and correction technique to eliminate transmission error.
- CO5: Analyze the fixed and variable length address (IPv4) subneting for the given scenarios.
- CO6: Understand the design issues of the transport layer.
- CO7: Understand the mechanism of protocols at application layer such as FTP, HTTP, Telnet, DNS.
- CO8: Understand IPv6 addressing and differentiate it from IPv4.

Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs):

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COs	POs/PSOs
CO1	PO1,PO3,PO12/PSO1
CO2	PO1/PSO2
CO3	PO1,PO4/PSO1,PSO4
CO4	PO1,PO3/PSO1
CO5	PO1,PO3,PO4,PO6/PSO3
CO6	PO2,PO4/PSO1
CO7	PO5,PO12/PSO2
CO8	PO4,PO7/PSO4

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BCSC0009: SOFTWARE ENGINEERING

Objective: *Be employed in industry, government, or entrepreneurial endeavors to demonstrate professional advancement through significant technical achievements and expanded leadership responsibility.*

L–T–P-J: 3–0–0-0

Credits: 03

Module No.	Content	Teaching Hours
I	**Introductory Concepts**: The evolving role of software – characteristics, components and applications. **Process Models:** Waterfall Model, Prototyping, Incremental, Spiral. **Agile software Development**: Introduction to Agile, Agile software development framework. **Software Requirement Specification**: Requirement Process, SRS Components, Requirement Specifications with Use Cases Diagram. **Software Project Planning:** Project Planning Objectives. **Software Metrics:** Size, Function Point, Staffing, Project Estimation Methods– COCOMO Model. **Function-Oriented Design**: Problem Partitioning, Abstraction, Top Down and Bottom Up Design. **Module-Level Concepts:** Coupling, Cohesion, Design Notation and Specification - Structure Charts; Structured Design Methodology - Data Flow Diagram, Sequence Diagram.	20
II	**OO Analysis and OO Design:** OO Concepts, Introduction to UML Design Patterns: Class Diagram, Activity Diagram, State Chart Diagram. **Coding**: Coding Process, Verification – Code Inspections, Software Metrics. **Testing Fundamentals**: Test Case Design, Black Box Testing Strategies, White Box Testing, Unit Testing, Integration Testing, System Testing. **Introduction to Automation Testing and Testing Tools:** Automated Testing Process, Framework for Automation Testing, Introduction to Automation Testing Tool. **Software Quality:** Models, ISO 9000 Certification for Software Industry, SEI Capability Maturity Model. **Software Maintenance**: Models Cost of Maintenance, Re-engineering, Reverse Engineering.	18

Text Books:

- R. S. Pressman , "Software Engineering: A Practitioners Approach", 7thEdition, McGraw Hill,2010.

Reference Books:

- K. K. Aggarwal and Yogesh Singh , "Software Engineering", 3rd Edition, New Age International Publishers,2008.
- Rajib Mall , "Fundamentals of Software Engineering", 3rd Edition, PHI Publication,2009.
- R.E Fairley , "Software Engineering", McGraw Hill,2004.
- Sommerville, "Software Engineering", 9th Edition, Pearson Education,2010.

Outcome: After the completion of the course, the student will be able to:

- CO1: Understand the basic concepts of software engineering.
- CO2: Apply software processes to solve real world problems.
- CO3: Estimate the cost, effort and schedule of software using COCOMO Model.
- CO4: Analyze the software design techniques (structure chart, SDM, sequence diagram).
- CO5: Understand the basic concepts of OO analysis and design.
- CO6: Develop the test cases to validate the software.
- CO7: Understand the basic models of software Quality and maintenance.

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Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs):

COs	POs/PSOs
CO1	PO1,PO7/PSO1
CO2	PO2,PO3/PSO4
CO3	PO2,PO11/PSO3
CO4	PO3,PO10/PSO4
CO5	PO3,PO7/PSO1
CO6	PO5,PO12/PSO2
CO7	PO4,PO9,PO12/PSO1

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BCSC1010: DISCRETE MATHEMATICS

Objective: *The objective is to introduce students to language and methods of the area of Discrete Mathematics. The focus of the module is on basic mathematical concepts in discrete mathematics and on applications of discrete mathematics in computer science.*

Credits: 4 **L–T–P-J: 3–1–0-0**

Module No.	Content	Teaching Hours
I	**Sets, Relations and Functions:** Introduction to Set Theory, Venn diagrams, algebra of Sets, Inclusion-Exclusion Principle, Partitions, Relations, Properties and their types, Function and their types. Recurrence Relations and Generating Functions **Introduction to Counting Principle**: Permutation, Combination, Permutation with Repetition, Combination with Repetition, Pigeonhole Principle. **Posets & Lattices:** Definition, Properties of lattices – Bounded, Complemented, Modular and Complete lattice. **Natural Numbers:** Introduction, Mathematical Induction, Variants of Induction, Induction with Nonzero Base cases. Proof Methods, Proof by counter – example, Proof by contradiction.	20
II	**Propositional Logic** - Logical Connectives, Truth Tables, Normal Forms (Conjunctive and Disjunctive), Validity; **Predicate Logic** - Quantifiers, Inference Theory **Algebra**: Motivation of Algebraic Structures, Finite Groups, Subgroups and Group Homomorphism; Lagrange's Theorem; Commutative Rings and Elementary Properties; **Graph Theory:** Trees: Definition, Binary tree, Binary tree traversal, Binary search tree. Introduction to Graphs, , Operations on Graphs, Representation of graphs, Types: Planner, Directed, Complete, Bipartite Graph, Isomorphism, Euler Graph, Hamiltonian Graph, Connectivity.	20

Text Book:

- Kenneth H Rosen , "Discrete Mathematics and Its Applications", 7th edition, TMH,2012.

Reference Books:

- J.P. Tremblay , "Discrete Mathematical Structures with Applications to Computer Science", TMH, New Delhi,1997.
- V. Krishnamurthy , "Combinatorics: Theory and Applications", East-West Press, New Delhi,1986.
- Ralph P. Grimaldi, "Discrete and Combinatorial Mathematics- An Applied Introduction", 5th Edition, Pearson Education,2004.
- C.L. Liu, "Elements of Discrete Mathematics", 2nd Edition, TMH,2000.

Outcome: After the completion of the course, the student will be able to:

- CO1: Understand the notion of mathematical thinking and proofs to solve the problem.
- CO2: Apply the basics of discrete probability and number theory to solve the real world problem.
- CO3: Analyze basic discrete structures and algorithms using effectively algebraic techniques.
- CO4: Analyze mathematical concepts like sets, reasoning, relational algebra and graph theory to solve optimization problems.
- CO5. Analyze the validity of an argument using logical notation.
- CO6. Demonstrate the basic structures of proof techniques to write and evaluate the validity of arguments.
- CO7. Understand the basic principles of sets, set equalities and operations in sets.
- CO8. Apply counting principles to determine probabilities.

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Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs):

COs	POs/PSOs
CO1	PO1,PO2/PSO1,PSO3
CO2	PO1,PO3/PSO4
CO3	PO2,PO3/PSO3
CO4	PO2,PO3/PSO3
CO5	PO1,PO2/ PSO3
CO6	PO1,PO3/PSO2,PSO3
CO7	PO1,PO2/PSO1
CO8	PO1,PO3/PSO1,PSO4

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BCSC0801: OBJECT ORIENTED PROGRAMMING LAB

Objective: *The objective of this course is that students will study and learn Object Oriented Modeling and programming.*

Credits:01 **L-T-P-J:0-0-2-0**

Module No.	Content	Teaching Hours
I& II	Programs in Java and python based on the concepts of: • Classes, Constructors, Polymorphism and Keyword Static. Programs based on the concepts of: • Inheritance, Multithreading Using Thread Class & Interface Runnable, String Handling, Generic Classes. Programs based on the concepts of: • Handling Database Connectivity. • Implementation of Collection Framework. Programs based on the concepts of: • Database Connectivity. • Retrieving Data from Database. • Parameters Passing, Execute many Method. • Cursor Attributes. • Invoke Stored Procedures. • Invoke Stored Functions.	24

Reference Books:

- Naughton, Schildt, "The Complete Reference JAVA2", 9th Edition, Oracle Press.
- Bhave&Patekar, "Programming with Java", Pearson Education
- Bret Slatkin: "Effective Python: 59 Specific ways to write better Python", Addison Wesley, 2015.

Outcome: After completion of course, the student will be able to:

- CO1: Implement object oriented language features.
- CO2: Design GUIs and Graphical programming.
- CO3: Design object oriented solutions for small systems involving database and event handling concepts.

Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs):

COs	POs/PSOs
CO1	PO1,PO2/PSO1
CO2	PO3,PO5/PSO2
CO3	PO3,PO5/PSO4

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BCSC0802: DATABASE MANAGEMENT SYSTEM LAB

Objective: *The lab aims to develop an understanding of different applications and constructs of SQL, PL/SQL.*

Credits:01 **L-T-P-J:0-0-2-0**

Module No.	Content	Teaching Hours
I & II	• Write the SQL queries for data definition and data manipulation language. • To implement various operations on a table. • To implement various functions in SQL. • To implement restrictions on the table. • To implement the concept of the grouping of Data. • To implement the concept of Joins in SQL. • To implement the concept of sub-queries. • To implement the concept of views, sequence. • To implement the concept of PL/SQL using a cursor. • To implement the concept of Procedure function and Triggers.	24

References Books:

- Date C J," An Introduction to Database Systems", 8th Edition, Addison Wesley.
- Korth, Silbertz and Sudarshan, "Database Concepts", 5th Edition, TMH, 1998.
- Majumdar& Bhattacharya, "Database Management System", TMH

Outcome: After the completion of the course, the student will be able to:

- CO1: Apply SQL queries for DML and DDL.
- CO2: Develop the SQL queries for real life scenarios.
- CO3: Implement the procedural language (PL/SQL) and Triggers.

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Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs):

COs	POs/PSOs
CO1	PO1,PO2/PSO1,PSO4
CO2	PO1,PO2/PSO1,PSO4
CO3	PO2,PO3,PO5/PSO2,PSO3

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BCSC0803: OPERATING SYSTEMS LAB

Objective: *The lab aims to develop understanding the operation of UNIX operating system.*

Credits:01 **L-T-P-J:0-0-2-0**

Module No.	Content	Teaching Hours
I & II	• Implement the following basic commands (with options) used in UNIX/LINUX OS. • Write and implement the basic vi editor commands. • Shell scripts that use simple commands. • Decision based Shell scripts. • Shell scripts related to strings. • Shell scripts using pipes. • Shell scripts with loop statements. • Demonstration and solution for race condition. • Demonstration and use of System Calls. • Implement the basics of IPC in UNIX.	24

Reference Books:

- SibsankarHalder and Alex a Aravind," Operating Systems", 6th Edition, Pearson Education, 2009.
- Harvey M Dietel, "An Introduction to Operating System", 2nd Edition, Pearson Education, 2002.
- D M Dhamdhere, "Operating Systems: A Concept Based Approach", 2nd Edition, 2006.
- M. J. Bach. , "Design of the Unix Operating System", PHI, 1986.

Outcome: After completion of course, the student will be able to:

- CO1: Implement the basic operations on UNIX operating systems.
- CO2: Demonstrate the working of systems calls.
- CO3: Demonstrate message passing in Unix operating system.

Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs):

COs	POs/PSOs
CO1	PO1,PO3,PO4/PSO1
CO2	PO1,PO2/PSO1
CO3	PO1,PO4,PO5/PSO1,PSO2

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BCSC0804: COMPUTER ORGANIZATION LAB

Objective: *The aim of the lab is to better understand the design of sequential Circuits such as Flip-Flops, Registers, and Counters.*

Credits: 01 **L–T–P-J: 0–0–2-0**

Module No.	Content	Lab Hours
I & II	• Bread Board Implementation of Flip-Flops. • Experiments with clocked Flip-Flops. • Design of Counters. • Bread Board implementation of Counters &Shift Registers. • Implementation of Arithmetic Algorithms. • Bread Board implementation of Adder/Subtraction (Half, Full). • Bread Board implementation of Binary Adder. • Bread Board implementation of Seven Segment Display. • Small Project based on combinational and sequential circuit.	24

Reference Books:

- D.W. Patterson , "Computer Organization and Design", 4th **Edition**, Elsevier Publication, 2008.
- William Stalling , "Computer Organization",8th Edition, PHI, 2011.
- M. Mano , "Computer System Architecture", 3rd Edition, PHI.

Outcome: After the completion of the course, the student will be able to:

- CO1: Implement the Combinational and Sequential Circuit.
- CO2: Demonstrate the working of counter and shift register.
- CO3: Demonstrate the working of ALU and seven segment displays.

Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs):

COs	POs/PSOs
CO1	PO2,PO3,PO5/PSO2
CO2	PO3,PO4/PSO2
CO3	PO3,PO5/PSO1,PSO2

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BCSC0805: DATA STRUCTURES & ALGORITHMS LAB

***Objective*:** *The objective of this course is that students will understand and implement simple data structures, able demonstrate different sorting and searching techniques. and will be familiar with graphs and their applications.*

Credits:01 **L-T-P-J:0-0-2-0**

Module	Content	Lab
I &II	• Program to implement various operations in a singly linked list. • Program to implement insertion, deletion and traversal in a doubly linked List. • Program to implement polynomial addition using linked list. • Program to demonstrate the various operations on stack. • Program to convert an infix expression into postfix expression. • Program to evaluate a given postfix expression. • Program to implement Tower of Hanoi problem using Recursion. • Program to demonstrate the implementation of various operations on linear and circular queue. • Program to demonstrate the implementation of insertion and traversals on a binary search tree. • Program to implement Dijkstra's Algorithm to find the shortest path between source and destination. • Program to search a given element as entered by the user using sequential and binary search to search a given element as entered by the user. • Implementation of various sorting algorithms like Selection Sort, Bubble Sort, Insertion Sort, Merge Sort, Quick Sort and Heap Sort.	24

Note: All Code must be done in Java as well as Python

Outcome: After completion of course, student will be able to:

- CO1: Demonstrate the associated operations in linear data structure like stack, Queue and link list.
- CO2: Demonstrate the associated operations in Binary Search Tree and Dijkstra's Algorithm.
- CO3: Implementation the sorting algorithms like Selection Sort, Bubble Sort, Insertion Sort, Merge Sort, Quick Sort and Heap Sort.

Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs):

COs	POs/PSOs
CO1	PO1/PSO1
CO2	PO4/PSO1,PSO3
CO3	PO2/PSO3,PSO4

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BCSC0806: MICROPROCESSORS LAB

Objective: *The objective is to introduce the Architecture and programming of the microprocessor and learning about interfacing and various applications of microprocessor.*

Credits: 01 **L–T–P-J: 0–0–2-0**

Module No.	Content	Lab Hours
I & II	• To study 8085 microprocessor System. • To study 8086 microprocessor System. • To develop and run basic programs in 8085 ALP. • To develop and run programs in 8085 ALP related to the concept of looping, counting and indexing. • To perform interfacing of RAM chip to 8085/8086. • To perform interfacing of keyboard controller. • To perform interfacing of DMA controller. • To perform interfacing of UART/USART.	24

Reference Books:

- Ramesh S. Gaonkar, "Microprocessor Architecture Programming and Applications with 8085", 4th Edition, Penram International Publishing, 2000.
- D. V. Hall , "Microprocessors and Interfacing: Programming and Hardware", 2nd Edition, TMH, 1992.

Outcome: After completion of course, student will be able to:

- CO1: Demonstrate the arithmetic and logical operations using assembly language programming (8085).
- CO2: Demonstrate the memory operations using assembly language programming (8085).
- CO3: Demonstrate the interfacing of Keyboard, DMA and UART controller.

Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs):

COs	POs/PSOs
CO1	PO1,PO3/PSO1,PSO2
CO2	PO1,PO2/PSO1,PSO2
CO3	PO1,PO3,PO5/ PSO2

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BCSE0105: MACHINE LEARNING

Objective: *To introduce students to the basic concepts and techniques of Machine Learning. To develop skills of using recent machine learning software for solving practical problems. To gain experience of doing independent study and research.*

Credits: 03 **L–T–P-J: 3–0–0-0**

Module No.	Content	Teaching Hours
I	**Introduction:** Machine Learning basics, Hypothesis space and inductive bias, training and test set, and cross-validation. **Introduction to Statistical Learning:** Bayesian Method. **Machine Learning:** Supervised (Regression, Classification) vs. Unsupervised (Clustering) Learning. **Data Preprocessing:** Imputation, Outlier management, One hot encoding, Dimensionality Reduction- feature extraction, Principal Component Analysis (PCA), Singular Value Decomposition **Supervised Learning:** Regression- Linear regression, Polynomial regression, Classification- Logistic regression, k-nearest neighbor classifier,	20
II	**Supervised Learning:** Decision tree classifier, Naïve Bayes classifier Support Vector Machine (SVM)Classifier, **Unsupervised Learning:** k-means clustering, Hierarchical clustering **Underfitting vs Overfitting:** Regularization and Bias/Variance. **Ensemble methods:** Bagging, Boosting, Improving classification with Ada-Boost algorithm.	20

Text Book:

- Tom M. Mitchell, Machine Learning. Tata McGraw-Hill Education, 2013.
- Alpaydin, E. . Introduction to machine learning. MIT press, 2009.

Reference Books:

- Harrington, P. , " Machine learning in action", Shelter Island, NY: Manning Publications Co, 2012.
- Bishop, C. M. . Pattern recognition and machine learning (information science and statistics) springer-verlag new york. Inc. Secaucus, NJ, USA. 2006

Outcome: After completion of Lab, student will be able to:

- CO1: Apply the basic concepts of machine learning.
- CO2: Apply the concepts of regression and re-sampling methods.
- CO3: Design supervised and re-enforcement learning based solution.
- CO4: Apply the ensemble methods for improving classification.
- CO5: Identify the ways of feature extraction, reduction and selection.
- CO6: Design the applications of machine learning algorithms.

Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs):

COs	POs/PSOs
CO1	PO1,PO2/PSO3,PSO4
CO2	PO1,PO2/PSO1,PSO3
CO3	PO1,PO3,PO5/PSO1,PSO3
CO4	PO1 /PSO1
CO5	PO2/PSO3
CO6	PO1,PO2,PO3/PSO1,PSO2,PSO4

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BCSE0133: MACHINE LEARNING LAB

Credits: 01 **L–T–P-J: 0–0-2-0**

S.No	PROGRAM
1	Estimate parameters of a model based on Linear Regression method using a given set of training data set.
2	Estimate parameters of a model based on maximum likelihood estimation method using a given set of training data set.
3	Compute weights of ANN based on back propagation method using a given training dataset.
4	Compute probability of a person to be diabetic based on a given dataset of diabetic persons using Naïve Bayesian classifier.
5	Classify a person as male or female based on a given dataset using naïve Bayesian Classifier, and calculate accuracy, precision, and recall for your data set.
6	Write a program to implement *k*-Nearest Neighbour method to classify the iris data set. Print both correct and wrong predictions. Use Java/Python ML library classes
7	Predicts whether the bank should approve the loan of an applicant, based on his profile using Ensemble learning method.
8	Apply Ensemble learning to cluster a set of data stored in a .CSV file. Use the same dataset for clustering using *k*-Means method. Compare the results of these two algorithms and comment on the quality of clustering. You can add Java/Python ML library classes/API in the program.
9	The stock prediction data is used to predict, whether the stock will go up or down. Perform the task of feature selection with the help of wrapper method.
10	Identify principal components of Big Mart sales data using Principal component analysis (PCA). Also plot the result of PCA, and give inferences.

Outcome: After studying the subject, the students will be able to:

- CO1: Applythe machine learning algorithms in the area of text, audio and image processing.
- CO2: Apply classification algorithms to design complex problems.
- CO3: Design solution to societal issues using machine learning algorithms.
- CO4: Analyze the view problem in the perspective of machine learning.

Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs):

CO	PO/PSO
CO1	PO3,PO5/PSO4
CO2	PO1,PO5/PSO1,PSO3,PSO4
CO3	PO3 /PSO1,PSO2,PSO4
CO4	PO2,PO3/PSO1,PSO2,PSO4

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